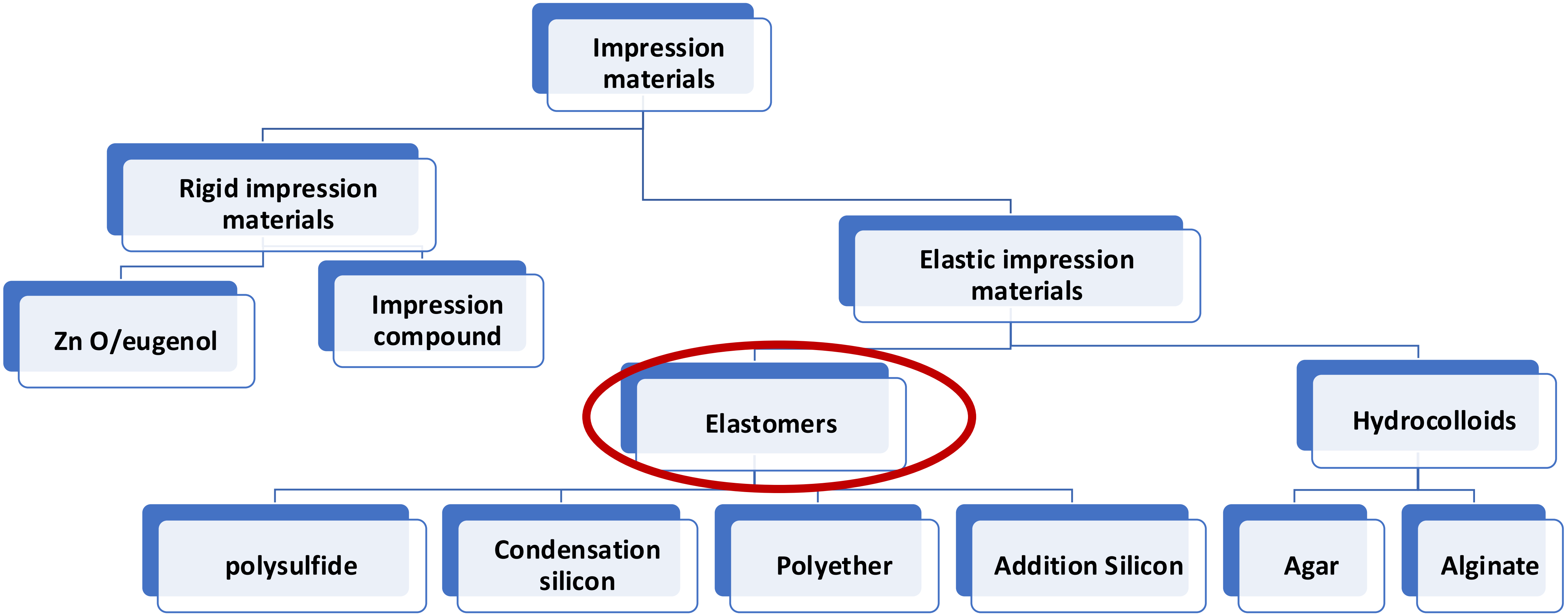
Elastic Impression materials II (Elastomers)

By
Dr. Sherif Refat Abd El Monem
Lecturer of Biomaterials

Classification of impression materials



Classification of Dental Impression Materials

Setting Mechanism	Rigid (inelastic)	elastic
Chemical reaction (irreversible)	Zinc oxide–eugenol	-Alginate -Polysulfide -Polyether -Condensation silicone -Addition silicone
Thermally induced physical reaction (reversible)	Impression compound	Agar

B- Elastomers

Introduction:

- **Elastomeric** impression materials when set become ***flexible cross-linked polymers.***

- Elastomeric impression materials dominate the market mainly Due to their **: 1-greater accuracy, 2-dimensional stability with time, and 3- ability to record details as compared with the hydrocolloid materials.**

- In this LECTURE we concentrate on the most commonly used elastomeric impression materials :
addition silicones and polyethers

Consistencies

extra low, low, medium, heavy, and putty, in increasing order of filler content

Extra-low and putty forms are only for condensation and addition silicones

According to ISO : ***type 0: putty,***
Type 1: heavy-bodied,
type 2: medium-bodied,
and type 3: light-bodied

Difference between different viscosities is related to **filler and polymer contents**

Putty has highest filler content and low polymer content so lowest polymerization shrinkage

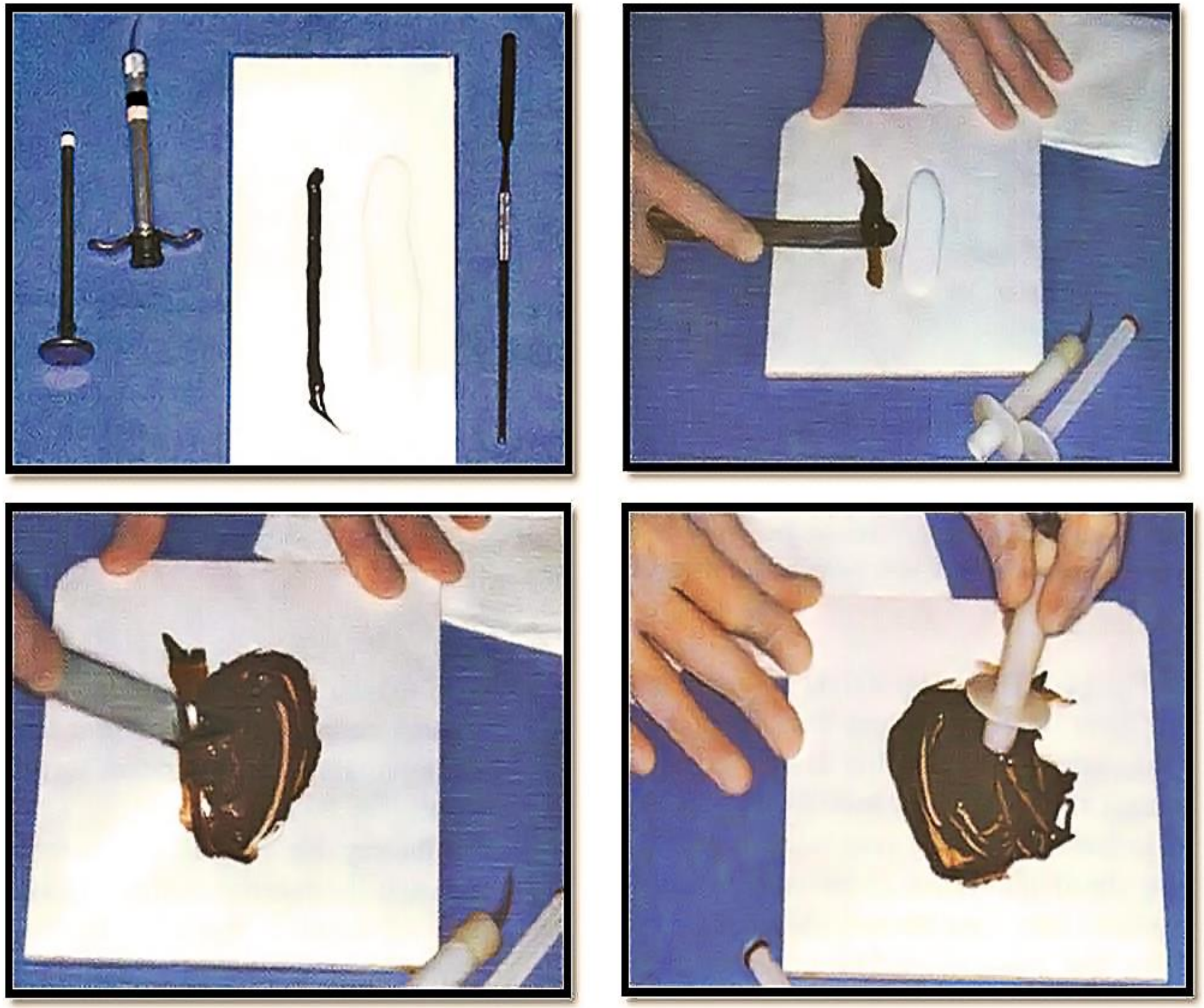
Mixing systems

Practical

- **<u>1- Hand mixing:</u>**

-the same lengths of materials onto a mixing pad
- The catalyst paste is first collected on a stainless steel spatula , then spread over the base paste
- Till uniform in color

- The two-putty systems for condensation and addition silicone are dispensed by an equal number of scoops of each material.



https://mydentaltechnologynotes.wordpress.com/2018/07/05/impression-materials-non-aqueous-elastomers/

Practical

- **2-Static *automixing:***
- The base and catalyst are in separate cylinders of the plastic cartridge. (all except putty)
- without mechanical mixing
- By flow division of streaming mix over alternating left and right turn helixes
- Uniform mix with no air-bubbles
- No motor driven

static *automixing system*

Practical

- **<u>3- Dynamic mechanical mixer :</u>**
- ease of use, speed, and good mix, but more expensive than hand and automixing

-The impression materials are supplied in **collapsible plastic bags placed in a Cartridge**

-uses a motor to drive parallel plungers, forcing the materials into a mixing tip to an impression tray or syringe.

- Motor driven mixing tip.

Practical

Mixing systems. A, Hand mixing. B, Static mixing. When the trigger is pulled, the plunger is driven forward (to the left), so that the base and catalyst pastes are forced from the cartridge into the mixing tip (*extreme left*).. C, Static mixing syringe.

D, Dynamic mechanical mixing.

https://pocketdentistry.com/8-impression-materials-2/

Practical

Impression Techniques

- **<u>Three common methods </u>**for making impressions for a fixed restoration are:**<u>1) a simultaneous, dual-viscosity technique,2) a single-viscosity or monophase technique, and 3) a putty-wash technique.</u>**

- **In general, Impression material is injected directly into the prepared teeth (low viscosity)and a tray containing the bulk of the impression material(higher viscosity) is placed thereafter.**

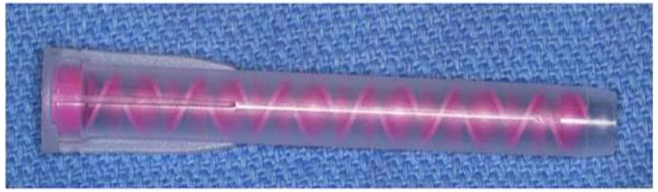
Practical

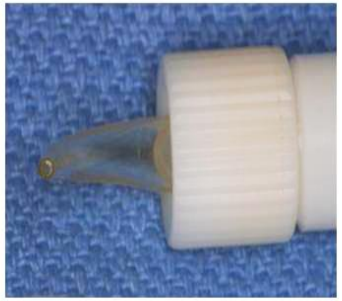
- **<u>1-Simultaneous, dual-viscosity technique:</u> low and high viscosity imp. Mat are mixed at the same time**

- **Low viscosity is injected by syringe on prepared teeth and critical areas**

- **High viscosity is placed in tray**

- **Then tray is placed in mouth**

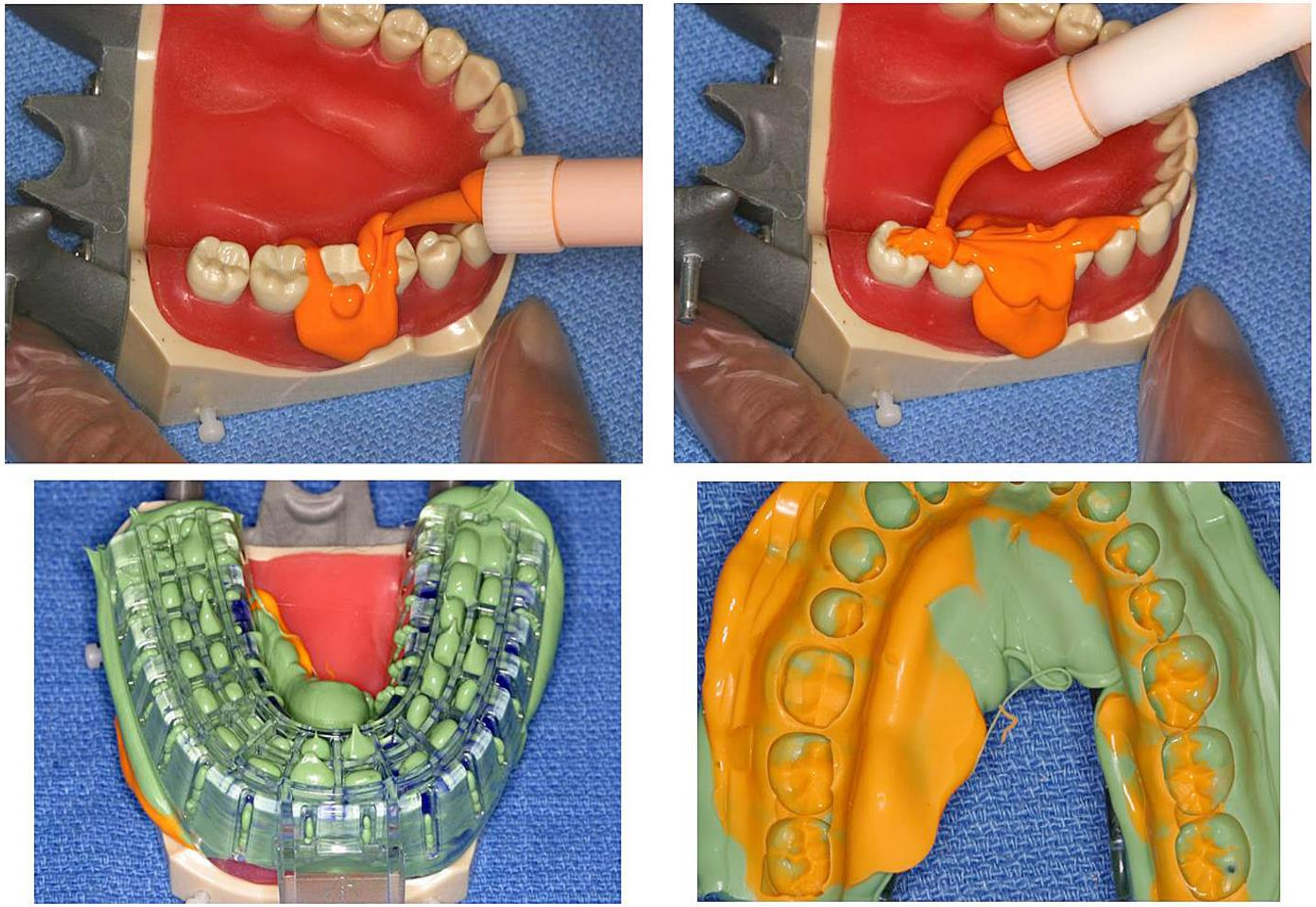




Auto-mixing of
light-bodied (syringe) material

Practical

Auto-mixing of
heavy bodied (tray) material



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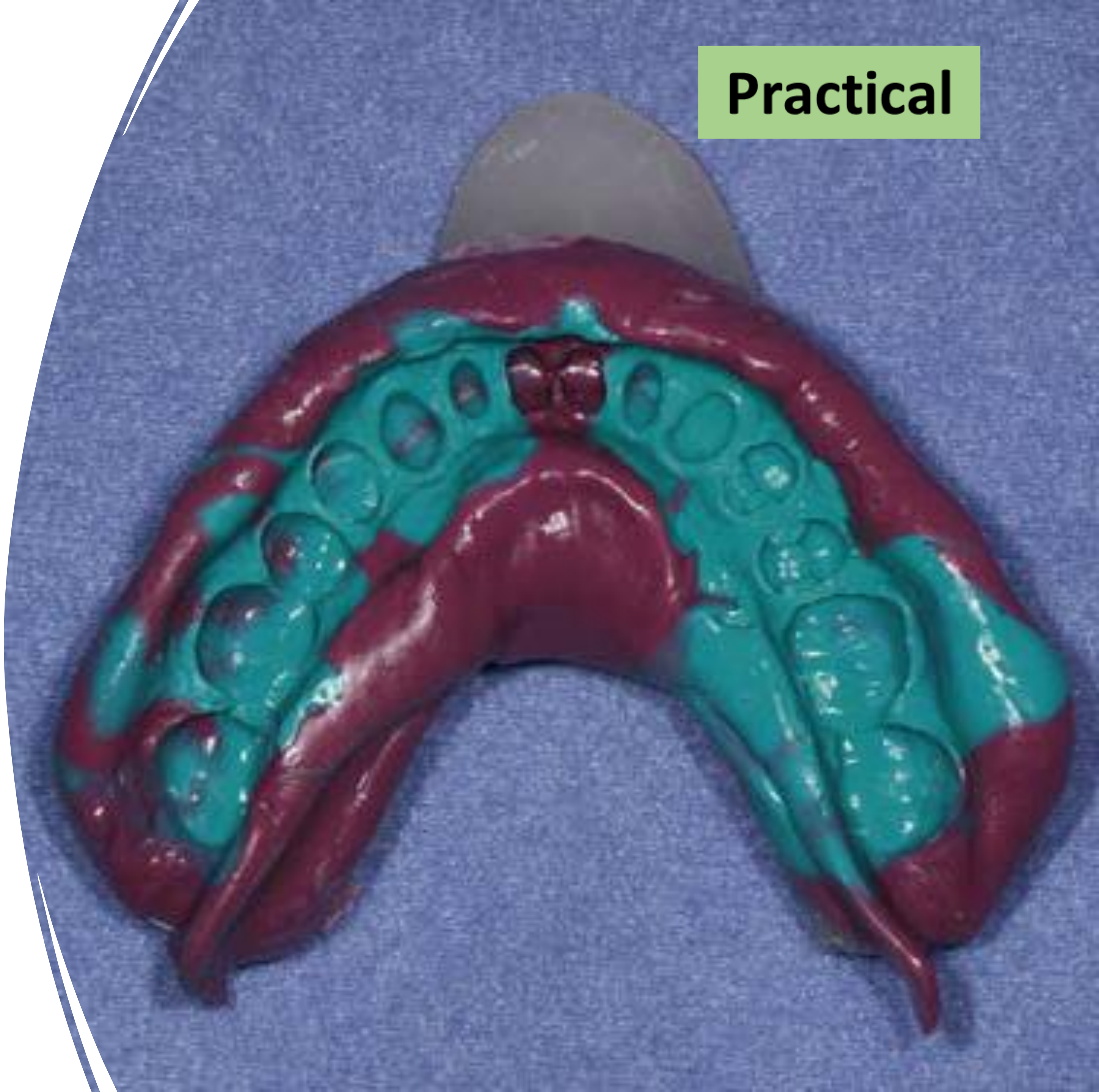
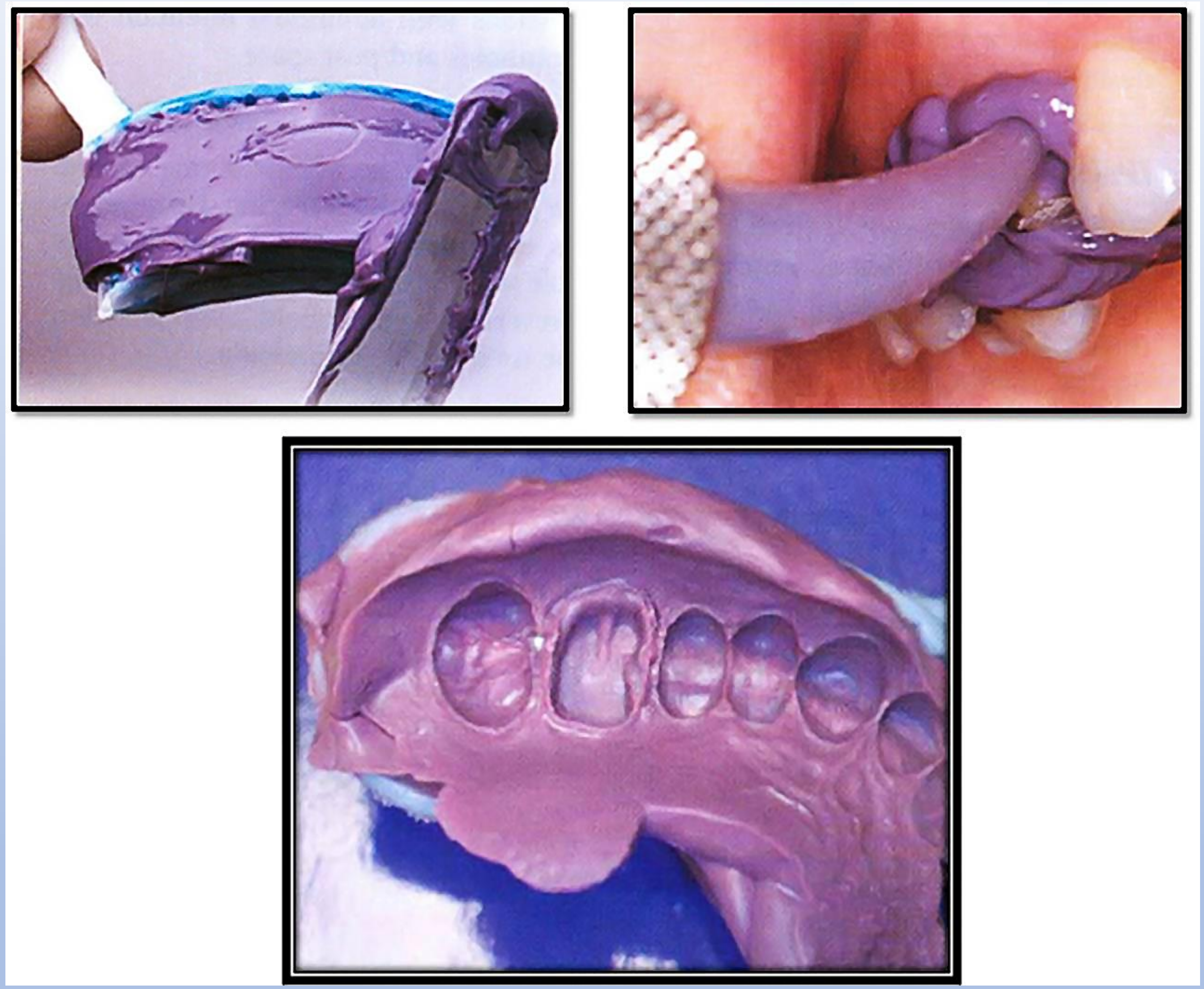


- An elastomeric addition-silicone impression.
- Turquoise material is of a low or injection consistency, and maroon material of a high or tray consistency. (*Courtesy Dr.*
- *Charles Mark Malloy, Portland, OR.*) (craig's textbook)

Practical

- **2- Single-viscosity or monophase** technique, impressions are taken with a medium-viscosity imp. Mat.

- **Pseudoplastic materials** :a decreased viscosity when subjected to high shear rates (shear thinning) e.g. addition silicon and polyethers

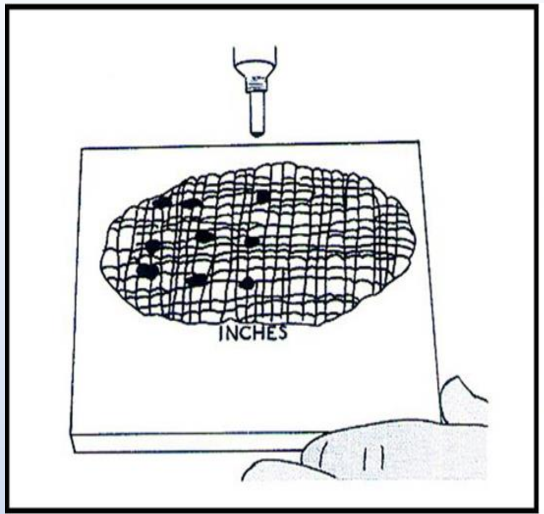


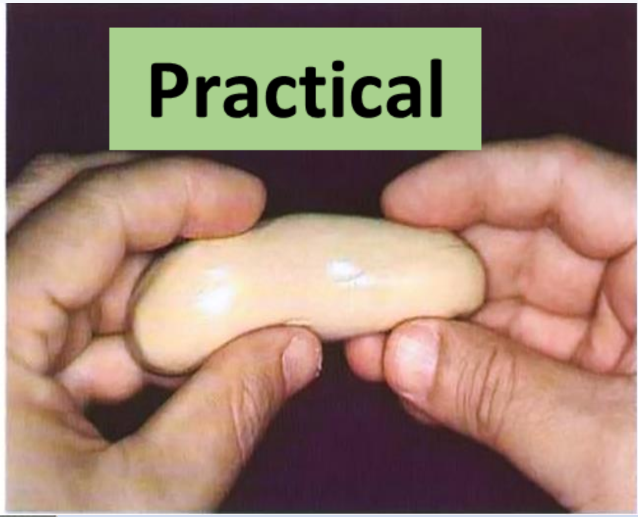
https://mydentaltechnologynotes.wordpress.com/2018/07/05/impression-materials-non-aqueous-elastomers/

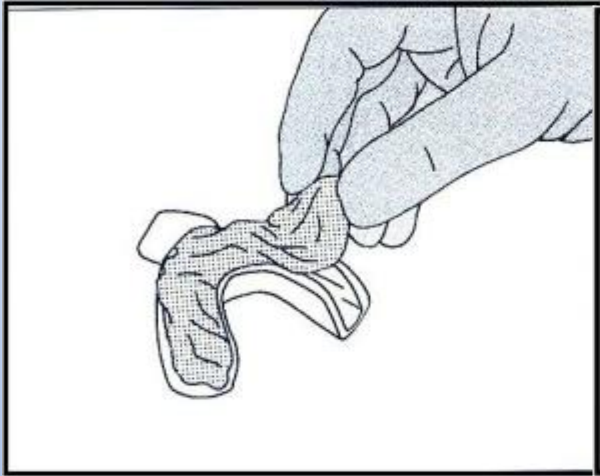
- 3- **<u>The putty-wash technique :</u>** a two-step impression procedure

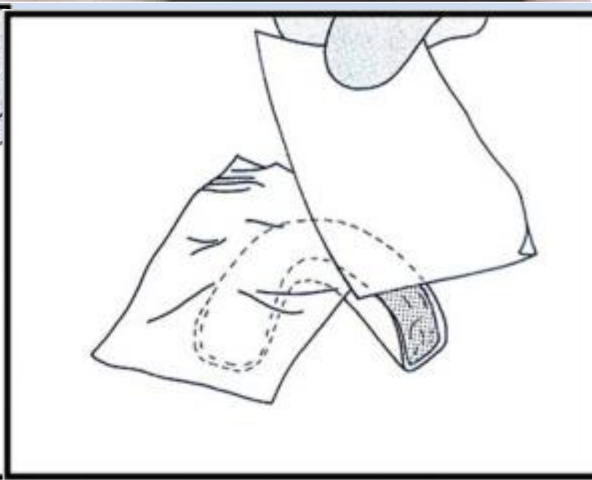
- a preliminary impression is taken in high- or putty-consistency material before preparation.

- Space is provided for a low consistency material, and after preparation, a low-consistency material is syringed into the area and the preliminary impression reinserted.









Practical

- General recommendations:

-Rigid stock tray can be used with addition silicon , condensation silicon and polyether

-Custom (special) tray must be used with polysulfides

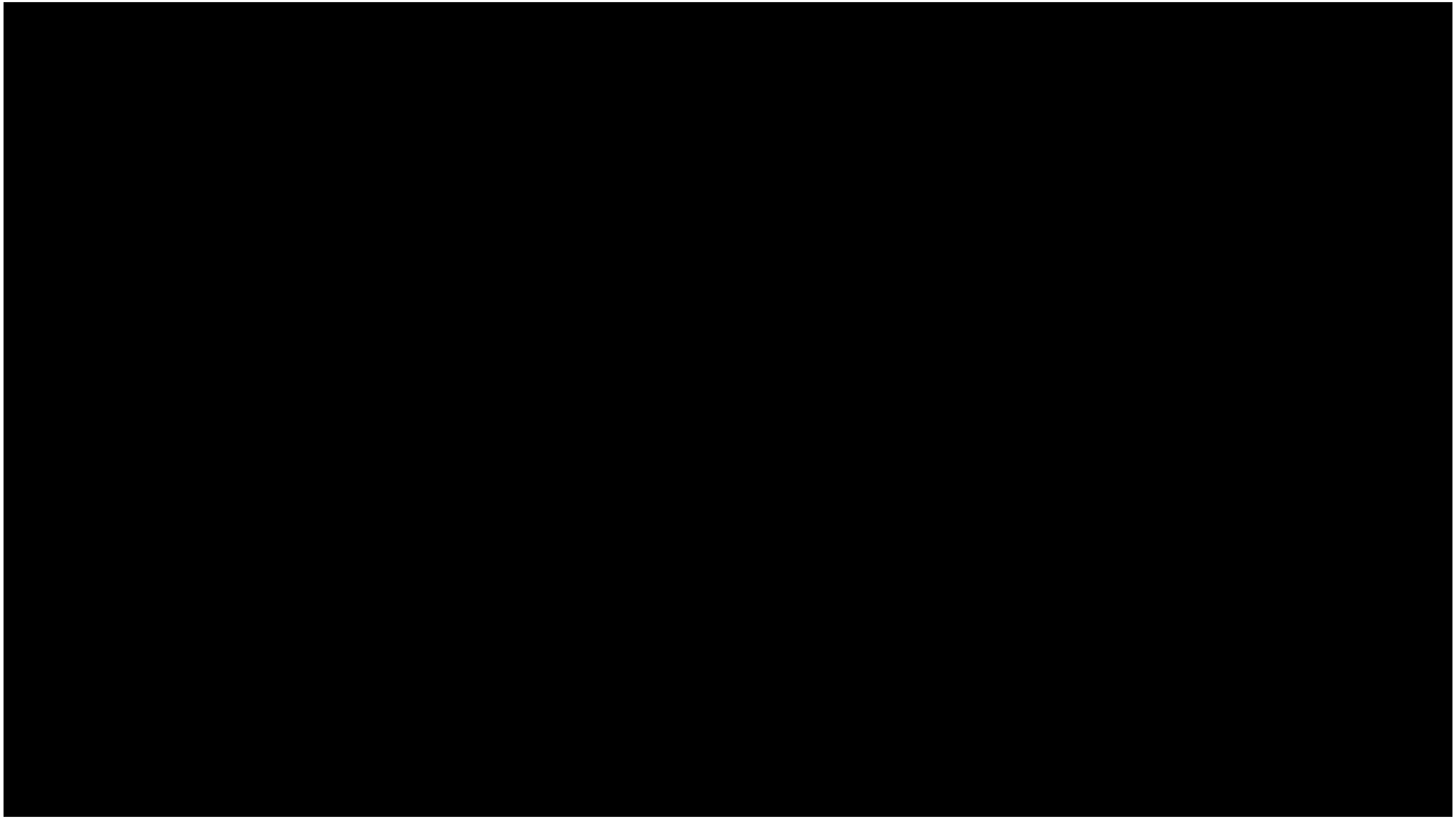
-Condensation silicon is used with putty wash technique

-Tray adhesive specific for each type is painted on tray before taking impression

Pocket dentistry website

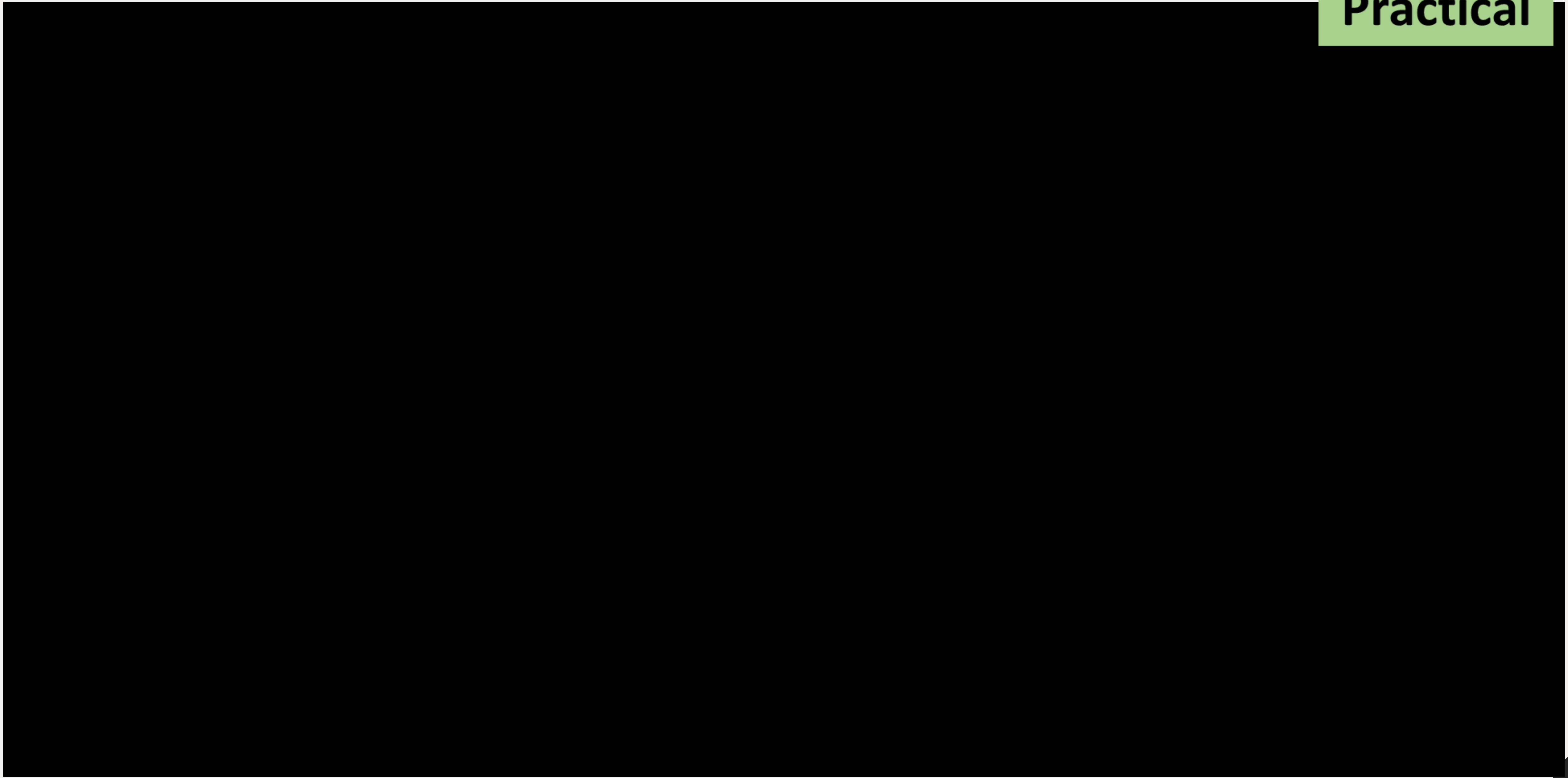
Practical

Practical



Practical

Practical



ADDITION SILICONE IMPRESSION MATERIALS

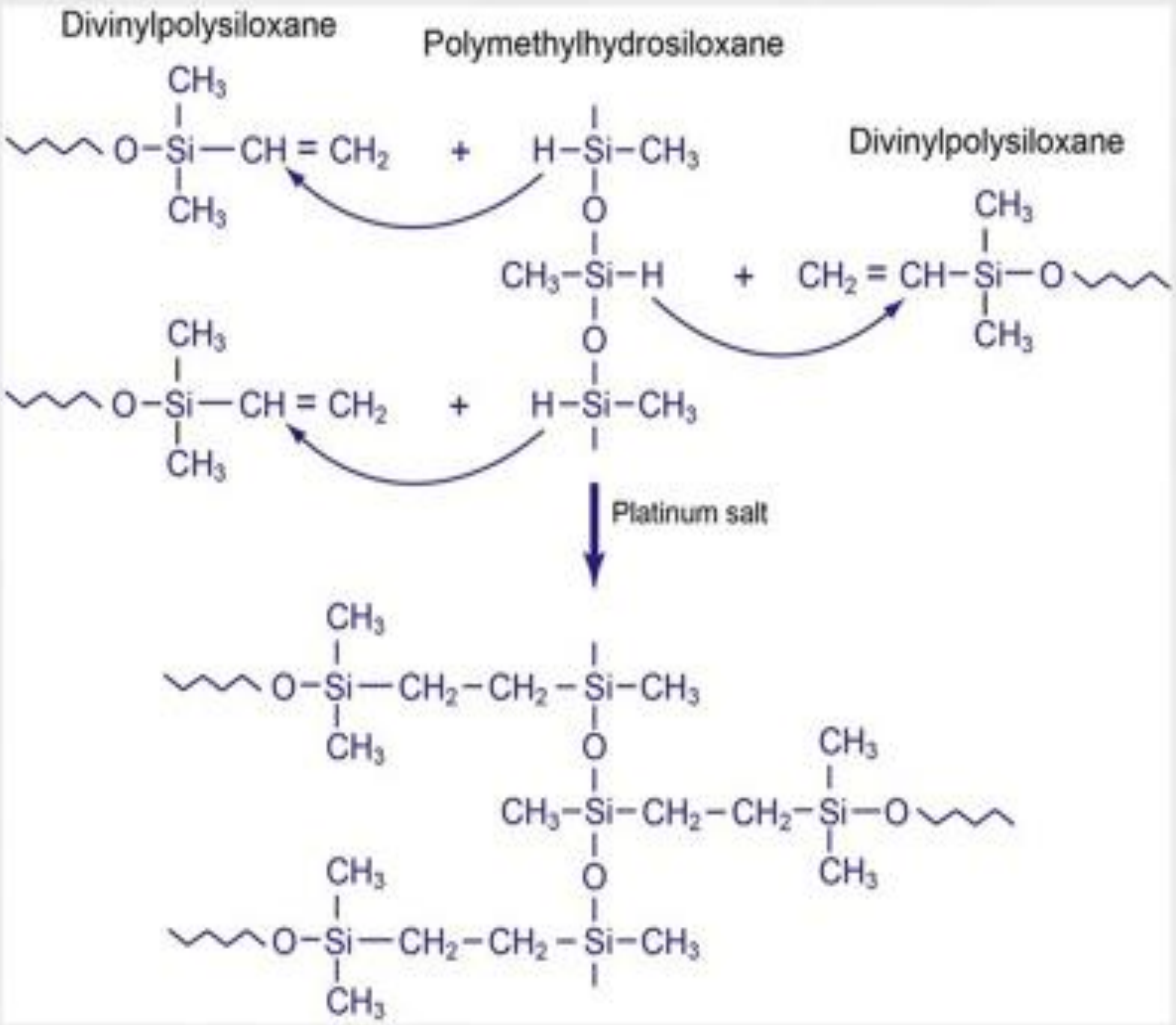
- **Addition silicone** impression materials (also known as vinyl polysiloxanes [VPS])
- **<u>Composition and Setting Reaction:</u>**
- The material is supplied as a **two-paste or a two-putty system**.
- One contains a **low-molecular-weight silicone with terminal vinyl groups, reinforcing filler, and a chloroplatinic acid catalyst.**
- and the other contains a **low-molecular-weight silicone with silane hydrogens and reinforcing filler.**

- **<u>A simplified setting reaction is as follows:</u>**
- Hydrogen-containing siloxane

+ Vinyl-terminal siloxane

+ Chloroplatinic ➡ Silicone rubber

- the addition reaction occurs between the vinyl and hydrogen groups with no by-product being formed



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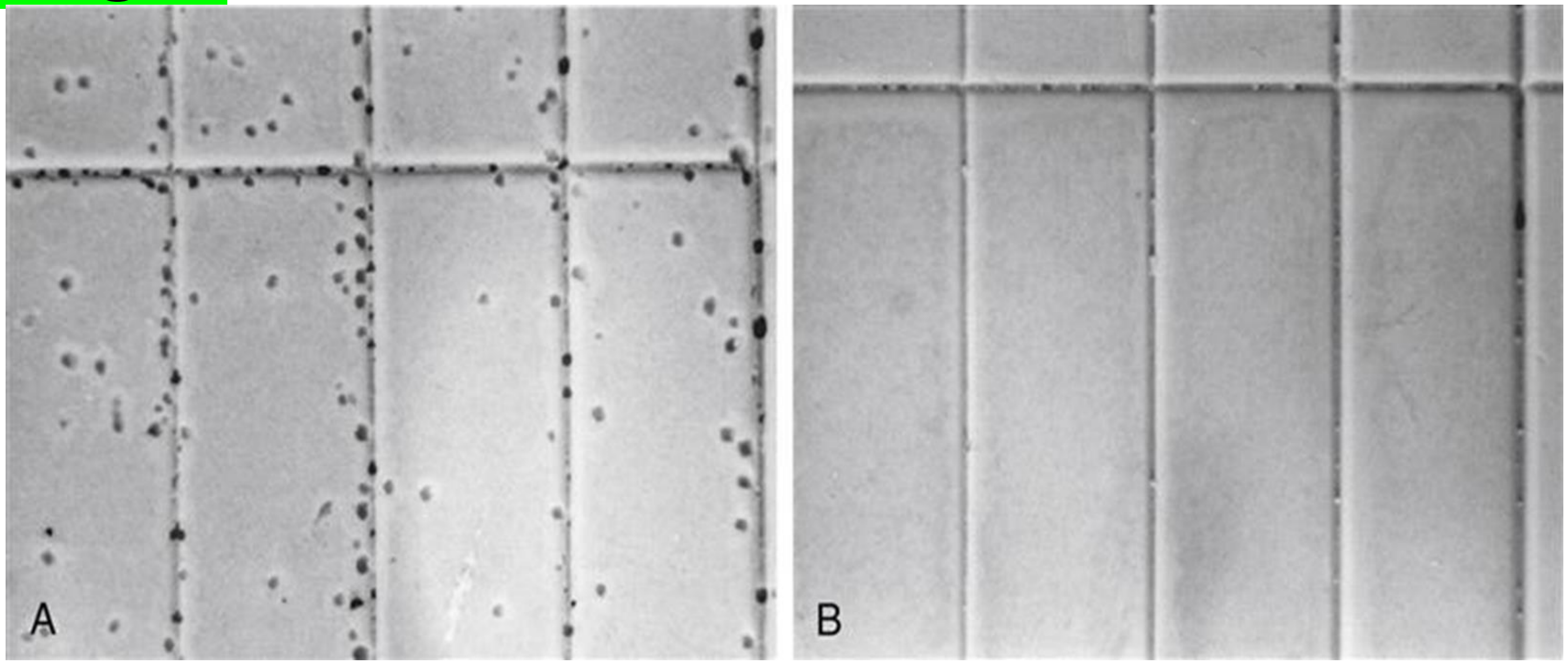
No volatile by-product (such as water or ethanol) is formed in this addition polymerization reaction ➡ minimal dimensional change occurs

- **If hydroxyl groups** are present in the addition silicone , that can lead to **hydrogen gas**.



https://pocketdentistry.com/12-replicating-materials-impression-and-casting/

The hydrogen can produce **bubbles** in gypsum dies or casts if poured in less than 30 minutes after the impression is taken.

wait at least 30 minutes for the setting reaction to be completed before the gypsum models and dies are poured..

Epoxy dies should not be poured until the impression has stood overnight

Some products allow immediate pouring by the inclusion of a hydrogen absorber, such as palladium. Therefore, the manufacturer's directions should be followed.

- **Latex gloves :** retard or stop the setting of addition-silicone impressions.
- **Why?** Due to sulfur contamination from latex gloves
- Do not touch prepared teeth or addition silicon with latex gloves
- **Vinyl and nitrile gloves** do not have such an effect (however some vinyl gloves may contain sulfur)

- **Properties:**
- The dimensional change in 24 hours is very low.
- The elastic recovery is about 99.8% (permanent deformation of 0.2%) which is the highest between all the impression materials.

- **High tear strength when compared to hydrocolloids**
- Elastomerics are **viscoelastic**

- **So, Snap rapid sharp removal of imp. is required**

- **Compatibility with gypsum products:** Nearly all addition silicones contain **surfactants,** so they are **hydrophilic** with contact angles of about 20° to 40°. Older types of addition silicon are hydrophobic and need spraying with surfactant before gypsum pouring.

C

Addition silicone impression materials are often supplied in cartridges of base and catalyst (A) that are auto-mixed through a dispensing tip and using a mixing gun (A, B). Alternatively, the material may be supplied as “sausages” (C) that fit into a power mixer. (Courtesy Y-W Chen, University of Washington Department of Restorative Dentistry, Seattle, WA.) (dental material foundations and applications)

Dental materials foundations and applications 11ed. ; Ch. Impression materials

- **DIMENSIONAL CHANGES ON SETTING :** 1-the major cause for **contraction** during setting is cross-linking and rearrangement of bonds between polymer chains (for both addition silicon and polyethers PE)

2-Addition-silicone & PE materials undergo relatively low **polymerization shrinkage**.

- **Dimensional stability :** 1- high when compared to hydrocolloids

3- The dimensional change in 24 hours of about - 0.1% “very low”

4- some types of addition silicon can be poured after 1 week

POLYETHER IMPRESSION MATERIALS (PE)

Polyether materials have properties similar to that of the addition silicones.

- The polyether polymer is **hydrophilic** and exhibits good wetting properties.

- Has high stiffness (low flexibility), some newer forms had improved flexibility by decreasing filler content.

The setting reaction mechanism is : **Cationic addition polymerization** by opening of the reactive terminal rings.

- **<u>Composition :</u>**
- **<u>base paste:</u>**- long-chain polyether copolymer with alternating oxygen atoms and methylene groups ($O–[CH_2]_n$) and reactive terminal groups

  - Silica fillers
  - plasticizers
  - triglycerides

- **<u>Catalyst paste :</u>** aliphatic cationic starter as a cross-linking agent.

  +

  silica filler and plasticizers

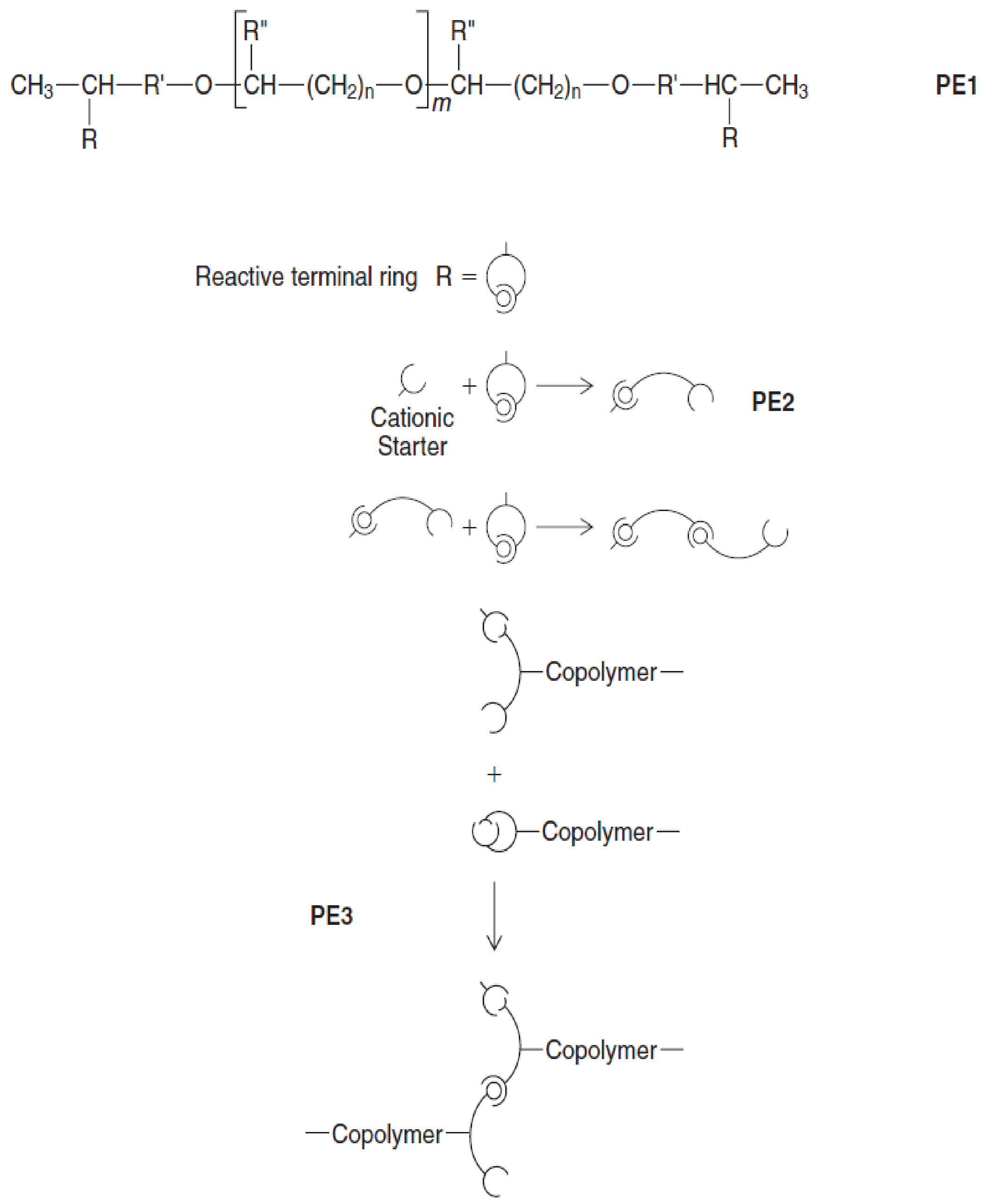
<u>The reaction mechanism is shown (PE2):</u>

Cationic polymerization by opening of the reactive terminal rings.

The backbone of the polymer is supposed to be a ***<u>copolymer of ethylene oxide and tetramethylene oxide units.</u>***

The reactive terminal rings open by the **<u>cationic initiator of the catalyst</u>**

reactive terminal ring becomes cation itself to open more rings then chain lengthening goes on (PE3)



Properties:

The elastic recovery percentage of the polyethers is slightly less than that of the addition silicones.

PE is viscoelastic so Sharp rapid removal of imp. From mouth is necessary

Properties

Higher tear strength than hydrocolloids

The polyether absorbs water and changes dimensions if stored in contact with water. So, polyether impressions should not be stored in water and must be washed and dried after removal from the mouth or disinfectants, disinfection not more than 10 min

- the polyether impression should be stored in a dry (relative humidity below 50%) away from sunlight, and cool environment to maintain its accuracy

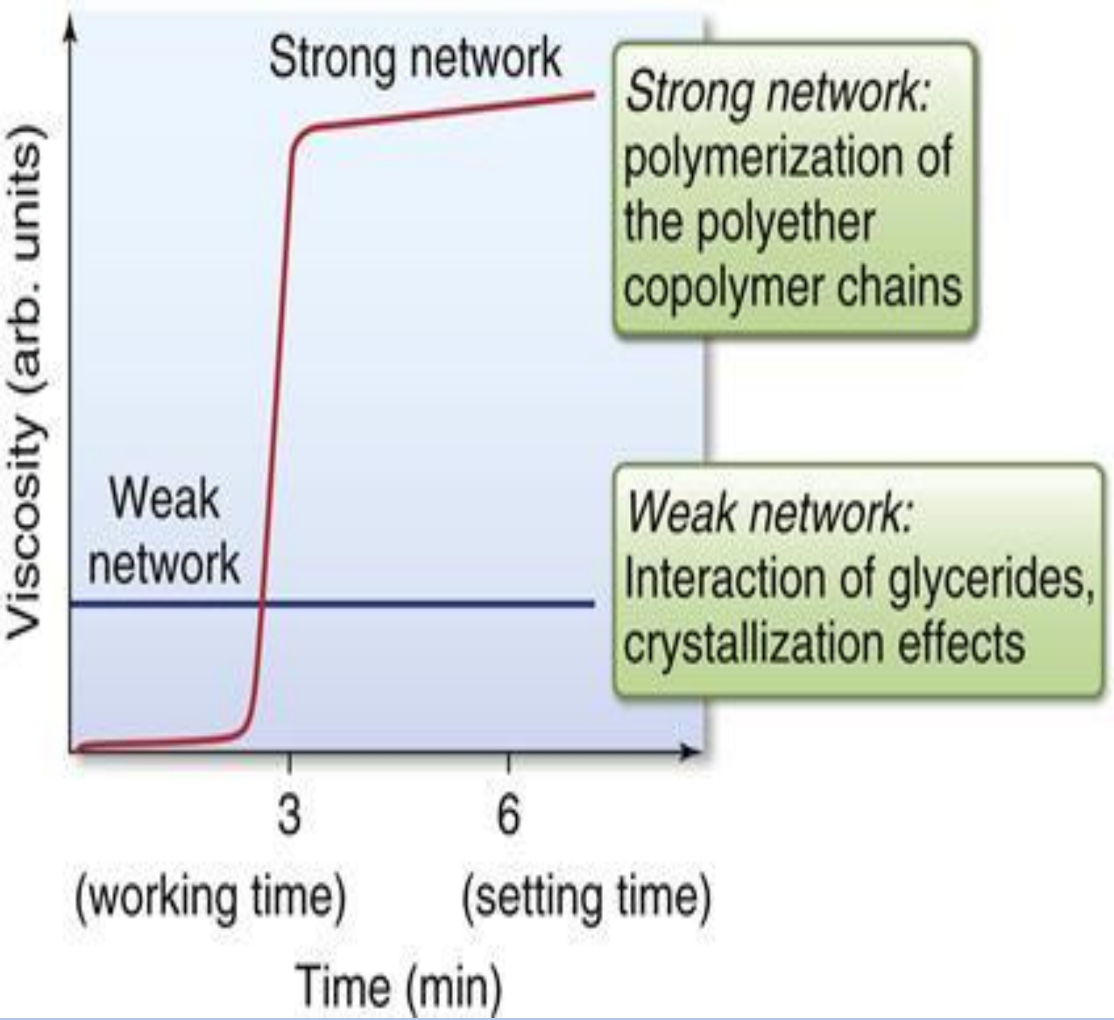
- **<u>Biocompatibility:</u>** The **aromatic sulfonic acid ester catalyst type** can cause skin irritation, and direct contact with the catalyst should be avoided.

- **Thorough and good mixing** of the catalyst with the base should be accomplished to prevent any irritation of the oral tissues.

Polyethers (PE) show defined working time with a sharp transition into the setting phase.

This behavior is often called *snap-set*.

Note : working and setting times of elastomeric impression materials are shortened by increases in temperature and humidity



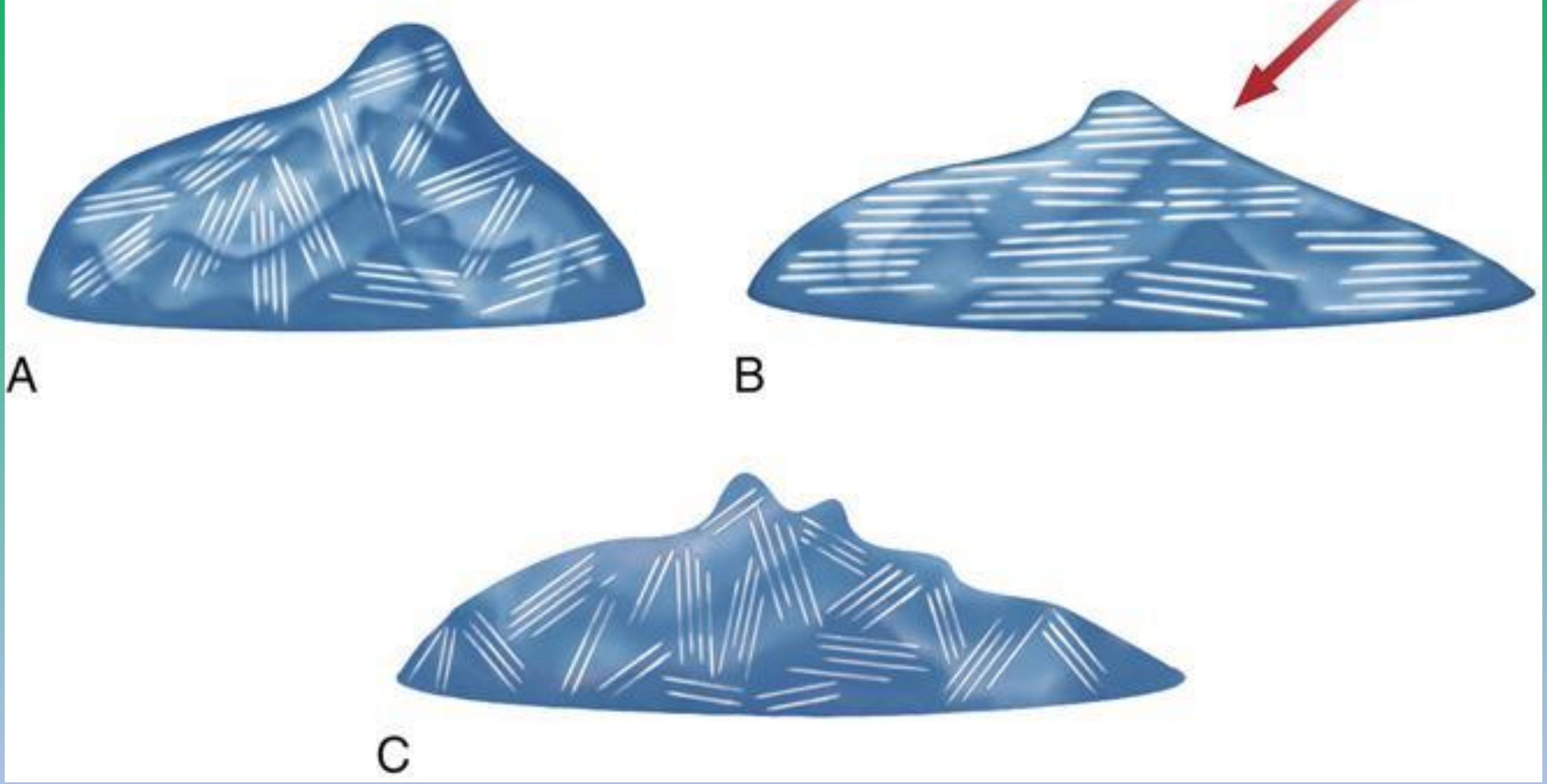
- **Polyethers are hydrophilic** , they are compatible with gypsum products, producing excellent cast surface.

- <u>**Viscosity :**</u> - **viscoelastic pseudoplastic properties** as addition silicones , which allows use of monophase impression materials

- **shear-thinning** due to a weak network of **triglyceride crystals in PE.**

- The crystals align when the impression material is sheared, but form a random network when not sheared.

https://www.researchgate.net/figure/Pour-the-cast-with-compatible-dental-plaster_fig1_282184121 https://pocketdentistry.com/12-replicating-materials-impression-and-casting/



Demonstration of the mechanism for the property of shear thinning or pseudoplasticity in polyethers. The trigliceride network, **A,** within the impression material aligns when sheared as with syringing, and **,** to achieve a lower viscosity **B**

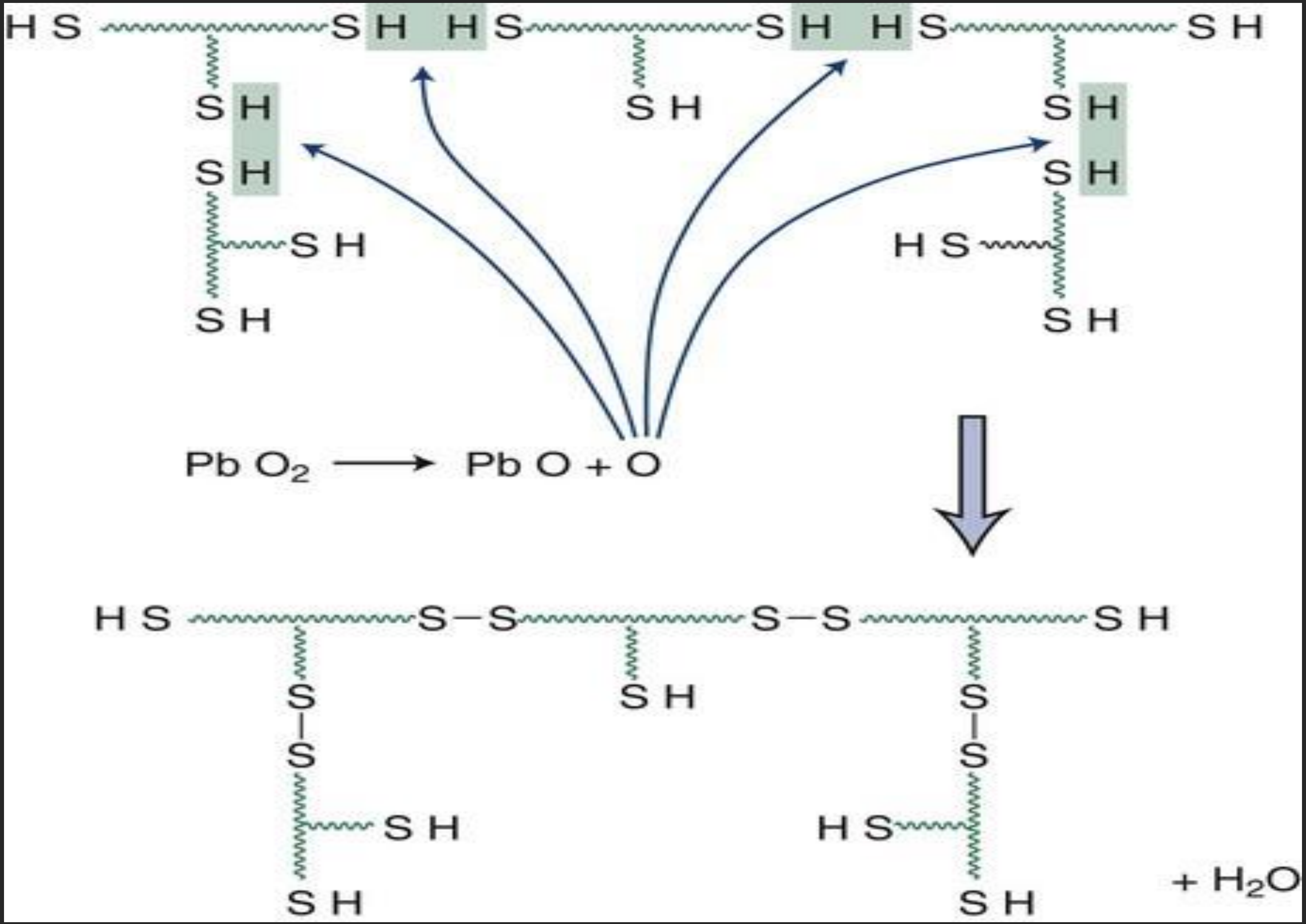
Once the shear force is removed, the viscosity increases with randomization of the triglyceride network, **C.**

The polyether materials are supplied in two mixing systems:
(1) as a cartridge with a mixing gun or (2) as a sausage with a dynamic mechanical mixer

The procedure is very simple: slide the PentaMatic™ foil bag into the cartridge and insert the cartridge into the Pentamix™ Mixing Unit. At the press of a button, the foil bag stretches into the opening of the integrated cartridge cap, where a spike locally ruptures the bag. The material can then flow out in a controlled manner.

POLYSULFIDE IMPRESSION MATERIALS

- The **polysulfide** impression materials (also called rubber base or mercaptan) are supplied as two pastes, with one tube labeled catalyst or accelerator and the other marked base.
- The **base material** consists of about 80% low-molecular weight organic polymer and contains reactive **mercaptan** groups (-SH) and 20% reinforcing agents
- The accelerator, or catalyst, tube contains a compound that causes the mercaptan groups to react to form a polysulfide elastomer. It contains ***lead dioxide, filler, plasticizer, and oleic or stearic acid*** as a retarder.


HS SH HS SH HS SH
SH SH SH SH
SH
HS
SH SH
$PbO_2 \longrightarrow PbO + O$
HS S–S S–S SH
S S
SH
S S
SH HS
SH SH
$+ H_2O$

- ***Three viscosities; light, regular and heavy***
- The most common catalyst is **lead dioxide**
- **Condensation polymerization , water by-product**
- The permanent deformation values of **2% to 3% (elastic recovery of 97% to 98%)**
- Polysulfide impression materials shrink **0.3% to 0.4%** during the first 24 hours
- Studies indicate that accuracy is improved if the models or dies are prepared **30 minutes after removal**, but no more long delays should occur.
- **lead-free** ***polysulphide*** is present
- the highest tear strength and lowest stiffness.

- Needs custom tray due to low stiffness

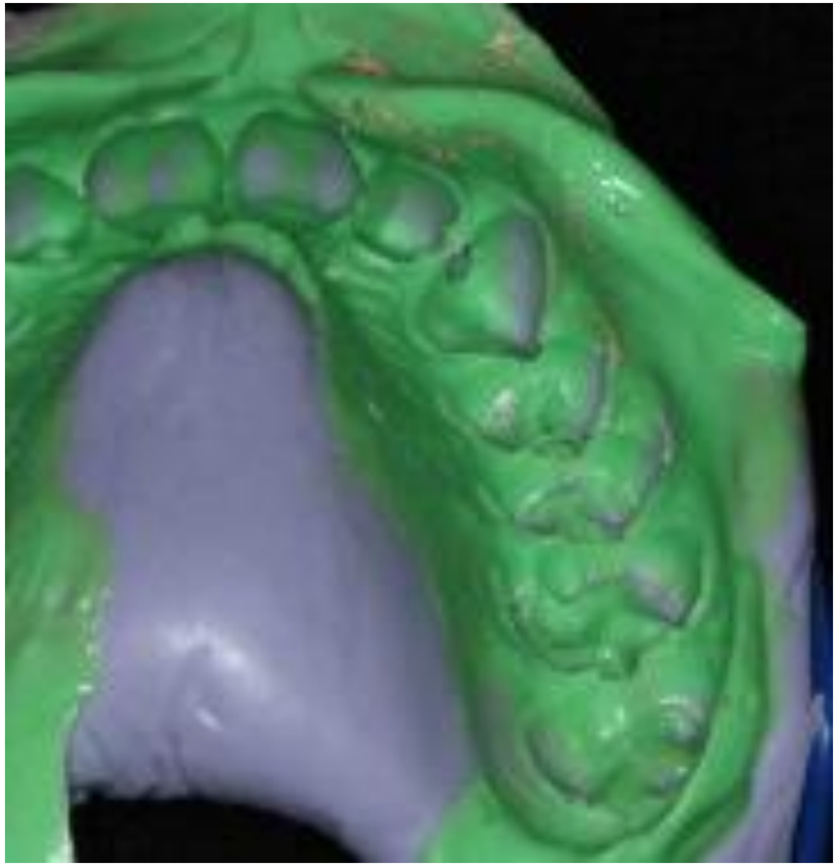
CONDENSATION SILICONE IMPRESSION MATERIALS

used as an accurate duplicating material in the dental laboratory.

The base paste contains a moderately low-molecular- weight silicone liquid, which has reactive –OH groups, Reinforcing agents such as silica

The accelerator consists of a tin organic ester suspension and an alkyl silicate.

- The **multifunctional ethyl silicate produces a network or cross-linked** structure.
- The **ethyl alcohol** produced as a by-product and gradually evaporates and cause high shrinkage during the first 24 hours after setting.
- **Putty-wash** technique is preferred
- The **dimensional change during 24 hours** after setting is relatively high.
- **About 50% of the dimensional change occurs during the first hour after setting, and the remaining 50% occurs between 1 and 24 hours**



Disinfection of elastomeric

- 1- follow general principles of Infection control IC , which were discussed before
- 2- disinfection by immersion could be followed for both addition silicon and polyethers (PE)
- 3- polyethers: disinfect by immersion with caution, short exposure disinfectant, not more than 10 minutes
- 4- addition silicones: disinfect by immersion, choose disinfectant conc. and type which needs no more than 30 minutes
- 5- Chlorine compounds and iodophors could be used for both polyethers and addition silicons.
- 6- Do not use glutaraldehydes

References

- **References and photos are from:**
- **1- Craig's 14th edition properties of restorative dental materials**
- **2- pocket dentistry website**
- **3- Philips science of dental materials**
- **4- text book of applied dental materials , faculty of dentistry , Cario university**

- THANK YOU